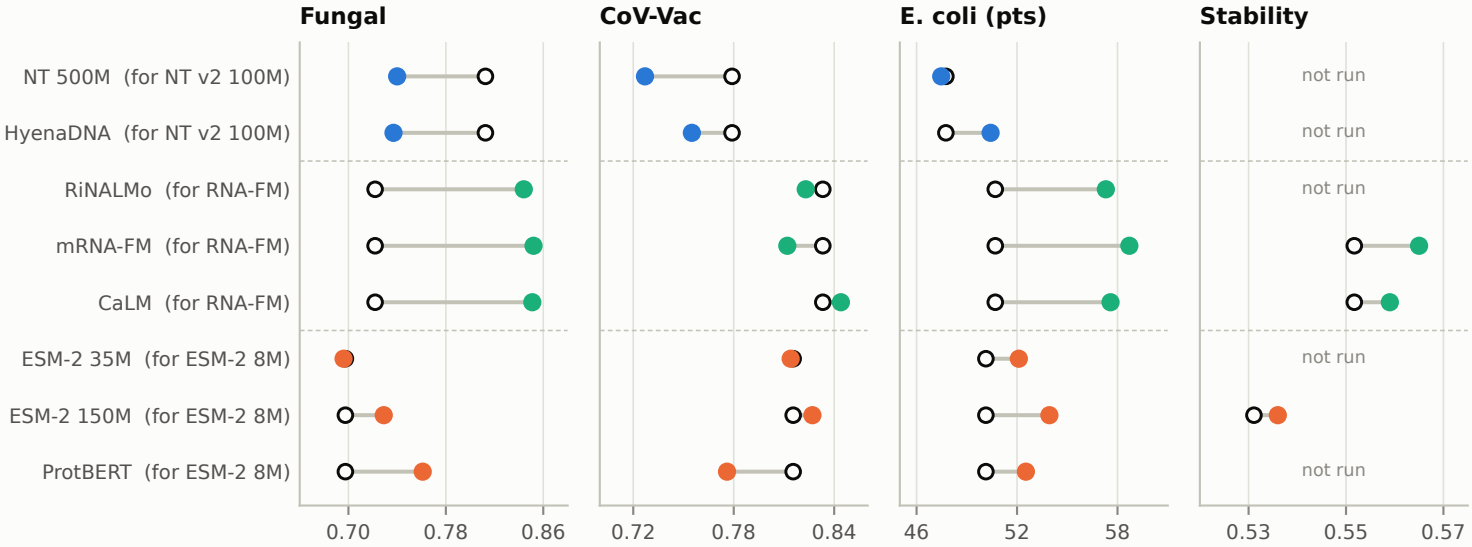
a Each swapped encoder read alone (filled) against the encoder it replaces (open)



b Fusion minus the best single encoder validation chose, one seed, 95% paired interval

	Fungal			CoV-Vac			E. coli (pts)			Stability		
	concat	gated	cross	concat	gated	cross	concat	gated	cross	concat	gated	cross
BioLangFusion's own	-0.020	-0.011	+0.011	-0.032	-0.050	-0.001	+10.5	+4.2	+3.7	+0.007	-0.001	+0.001
NT 500M	+0.000	-0.003	+0.016	**-0.080**	**-0.123**	-0.022	**+7.7**	**+4.1**	**+5.2**	not run		
HyenaDNA	-0.007	-0.014	+0.010	-0.001	**-0.036**	**+0.039**	**+7.9**	**+2.9**	+1.8	not run		
RiNALMo	+0.002	-0.013	+0.003	-0.017	-0.004	**+0.031**	**+3.9**	**+6.5**	-1.7	not run		
mRNA-FM	+0.001	-0.013	-0.012	**-0.054**	**-0.050**	+0.009	+0.3	**-5.6**	-3.9	+0.003	-0.000	-0.003
CaLM	-0.003	-0.003	+0.006	**-0.027**	**-0.042**	-0.017	+0.1	-0.8	**-3.7**	+0.002	+0.002	-0.010
ESM-2 35M	-0.018	-0.033	+0.006	-0.025	**-0.040**	+0.018	**+5.6**	+1.1	+1.1	not run		
ESM-2 150M	-0.028	-0.028	-0.002	**-0.031**	**-0.050**	+0.020	**+3.8**	+1.5	+0.7	+0.008	-0.003	+0.007
ProtBERT	**+0.044†**	**+0.039†**	**+0.054†**	**-0.045**	**-0.043**	+0.010	+1.7	**+3.9**	+2.1	not run		

fusion ahead, resolved · fusion behind, resolved · tie: interval crosses zero · original encoders, seed 0, for reference